Mechanism Questions #4

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This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*.

Explain the reaction mechanisms for the transformations below. Draw out each step and show electron movement using curved arrows. Identify the likely rate-determining step and identify the driving force for each reaction.

Each reaction is a single step in a multistep synthesis described in the referenced literature. For extra experience, access each paper and describe every step in the synthesis that is presented.

This is our final organic review set and so it is large. Keep going as we move forward to other topics. You can never practice to much.

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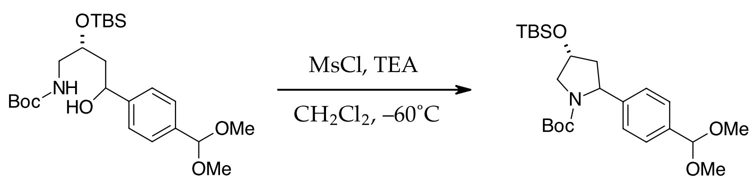

"Stereoselective Synthesis of a Broad Spectrum 1β-Methylcarbapenem, J-114,870." H. Imamura et al., *Tetrahedron*, **2000**, *56*, 7705-7713. https://doi.org/10.1016/S0040-4020(00)00685-2

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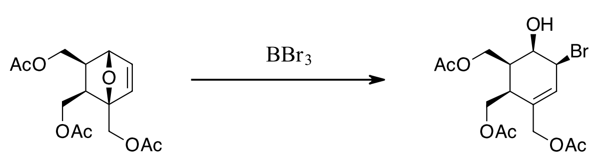

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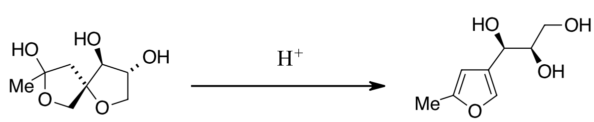

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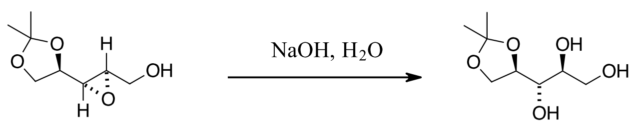

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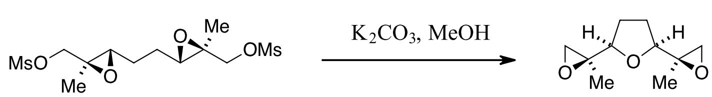

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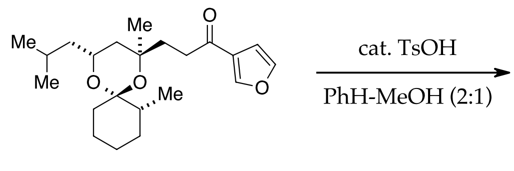

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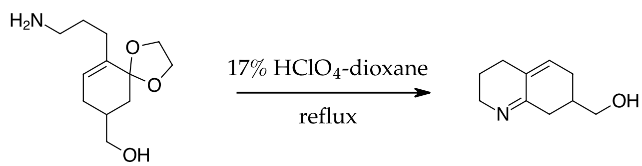

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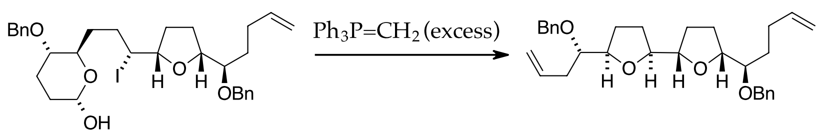

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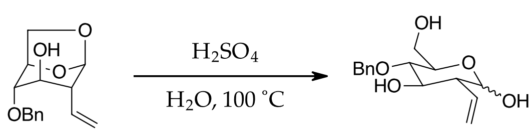

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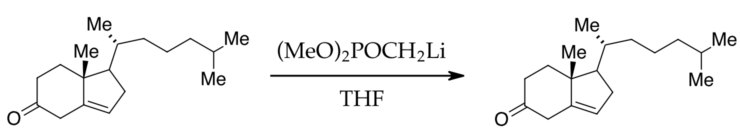

"Synthesis of a Vitamin D_3 Hydrindan Ring–Side-Chain Building Block, Involving Tandem Conjugate Addition and Alkylation Reactions." P. Grzywacz et al., *J. Org. Chem.*, **1997**, *62*, 5293–5298. https://doi.org/10.1021/jo9705834

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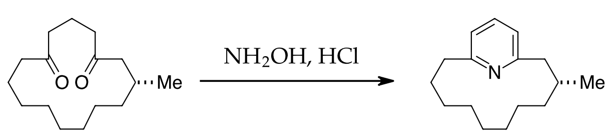

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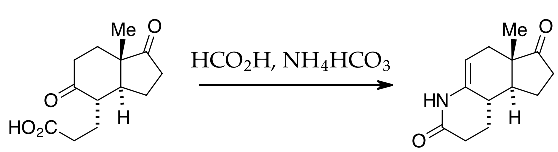

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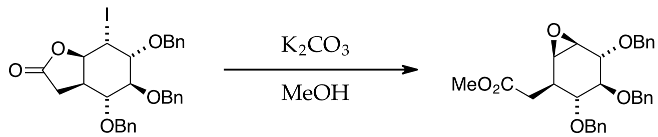

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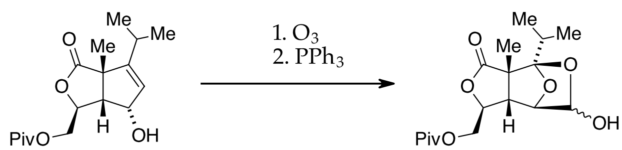

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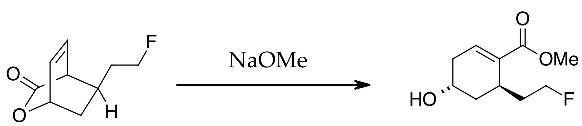

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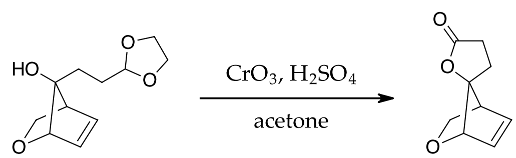

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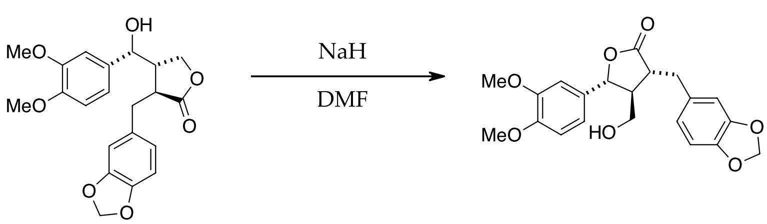

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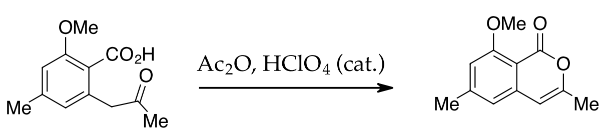

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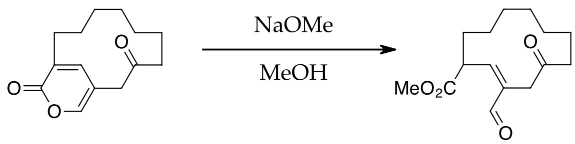

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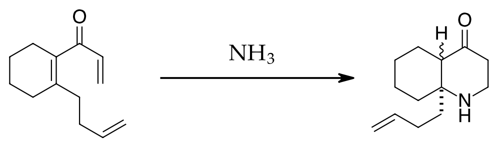

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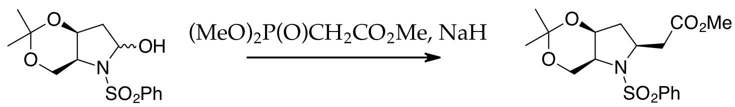

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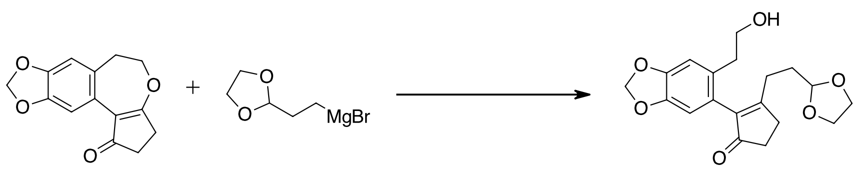

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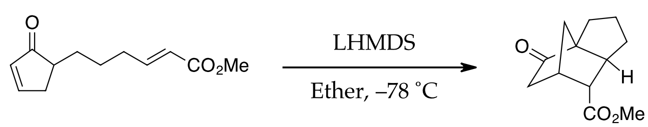

"Facile Construction of the Tricyclo[5.2.1.$0^{1,5}$]decane Ring System by Intramolecular Double Michael Reaction: Highly Stereocontrolled Total Synthesis of (±)-8,14-Cedranediol and (±)-8,14-Cedranoxide." M. Ihara et al., *J. Org. Chem.*, **1999**, *64*, 1259-1264. https://doi.org/10.1021/jo981996n

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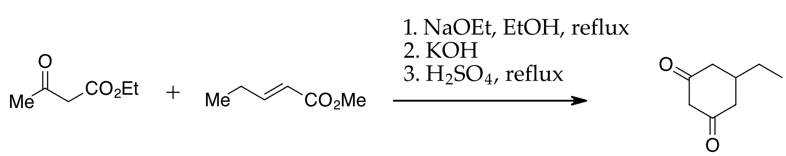

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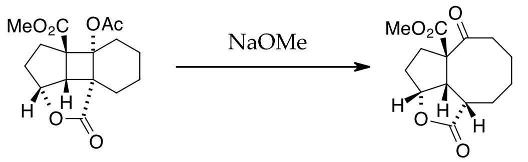

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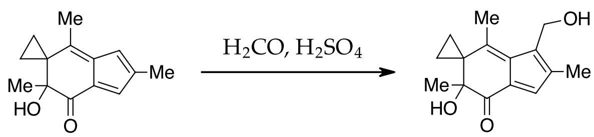

"A Rapid Synthesis of Hydroxymethylacylfulvene (HMAF) Using the Allenic Pauson-Khand Reaction. A Synthetic Approach to Either Enantiomer of This Illudane Structure." K.M. Brummond et al., *J. Am. Chem. Soc.*, **2000**, *122*, 4915-4920. https://doi.org/10.1021/ja000546l

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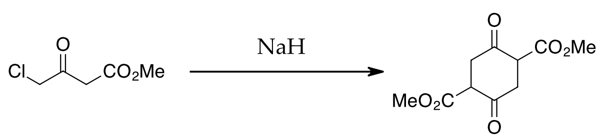

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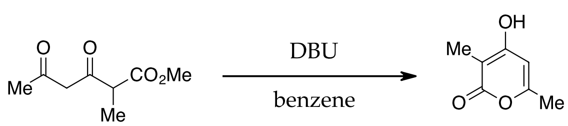

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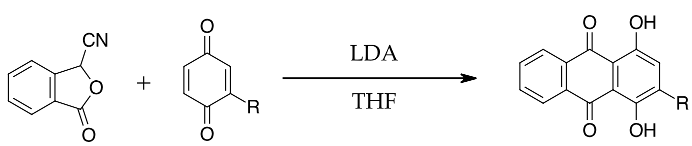

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Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will have…

- reviewed more organic chemistry mechanisms and are able to describe reactions that have been seen again and again.
- developed a vocabulary for chemical mechanism and are able to describe mechanism using words as well as with diagrams.